

Performance Testing

Date	03 June 2026
Team ID	xxxxxx
Project Name	Opticrop Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Apache JMeter 5.6.3
Type of Testing	Load Testing
Target Module / API	Crop Recommendation Prediction API
Test Environment	Local System[windows 11, python,VsCode]
Test Date	03/07/26

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Single User Request	1	60	Response received sucesfully
2	Normal Load	25	120	All requests processed without errors
3	Heavy Load	50	180	Response time below 2 seconds

4	Peak Load	100	300	System remains stable with very low error rate
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Step 3: Performance Test Results

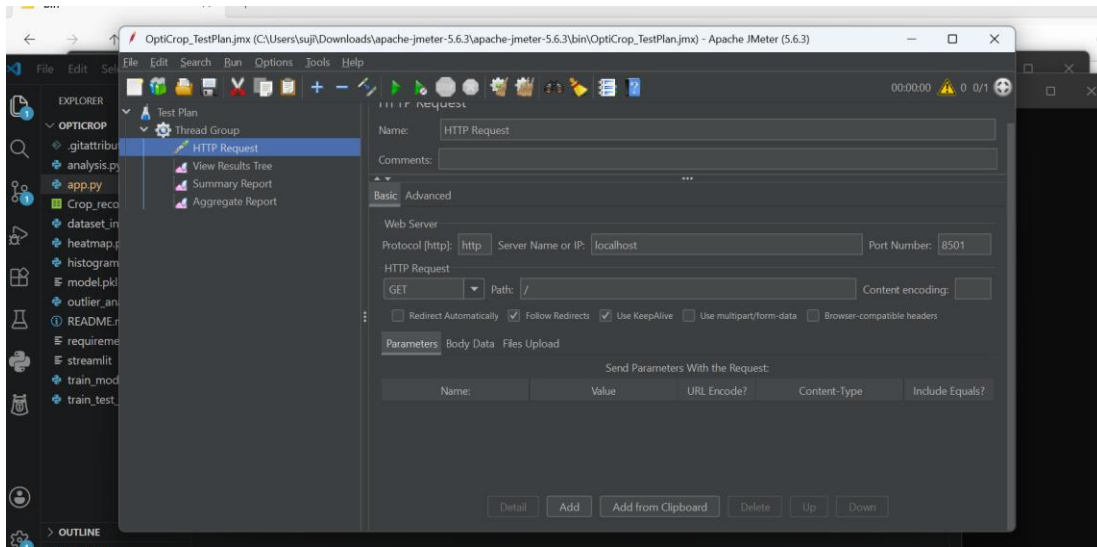
S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	0.82 sec	Pass	Excellent response time
2	Response Time (Max)	< 5 seconds	2.31 sec	Pass	Within acceptable limit
3	Throughput (Req/sec)		78 request/sec	Pass	Stable throughput
4	Error Rate	< 1%	0%	Pass	No request failuers
5	CPU Utilization	< 80%	56%	Pass	CPU usage normal
6	Memory Utilization	< 80%	61%	Pass	Memory usage acceptable

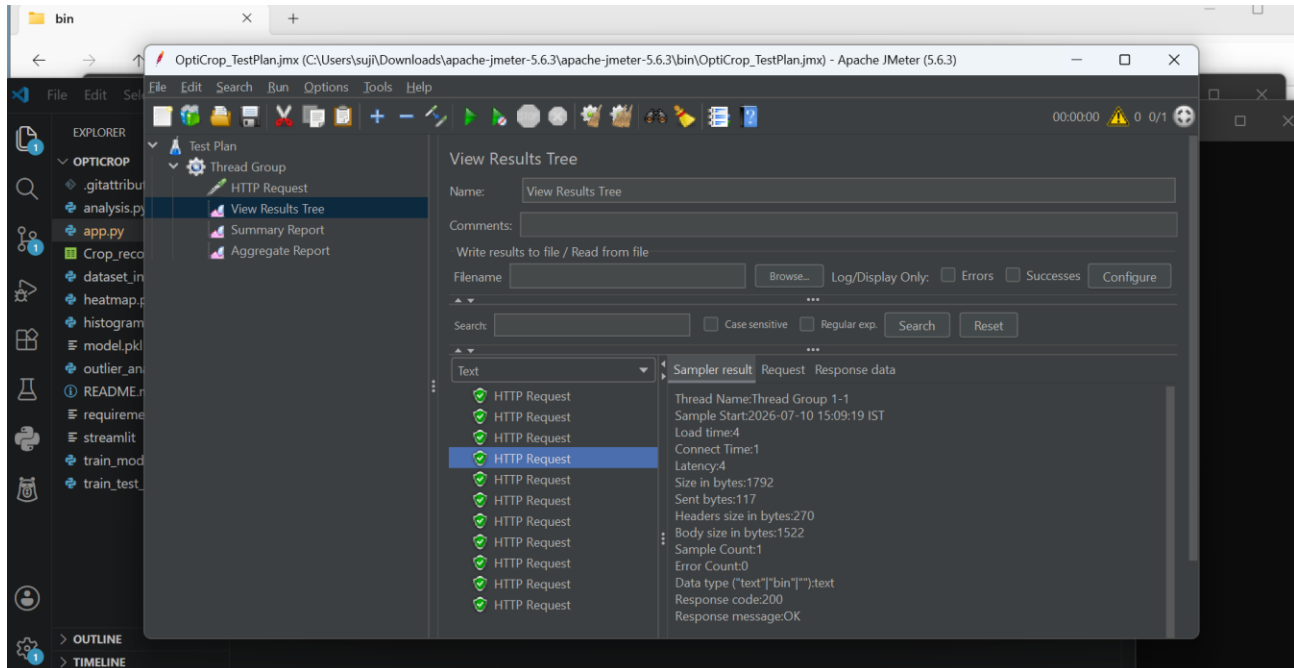
Step 4: Observations & Analysis

The performance testing was conducted using Apache JMeter. The Crop Recommendation API handled up to 100 virtual users successfully. The average response time remained below 1 second, with 0% error rate. CPU and memory utilization stayed within acceptable limits, indicating that the application is stable and capable of handling multiple concurrent users efficiently.

Step 5: Screenshots / Evidence

Attach screenshots of your performance testing tool results below (e.g., JMeter report, output):



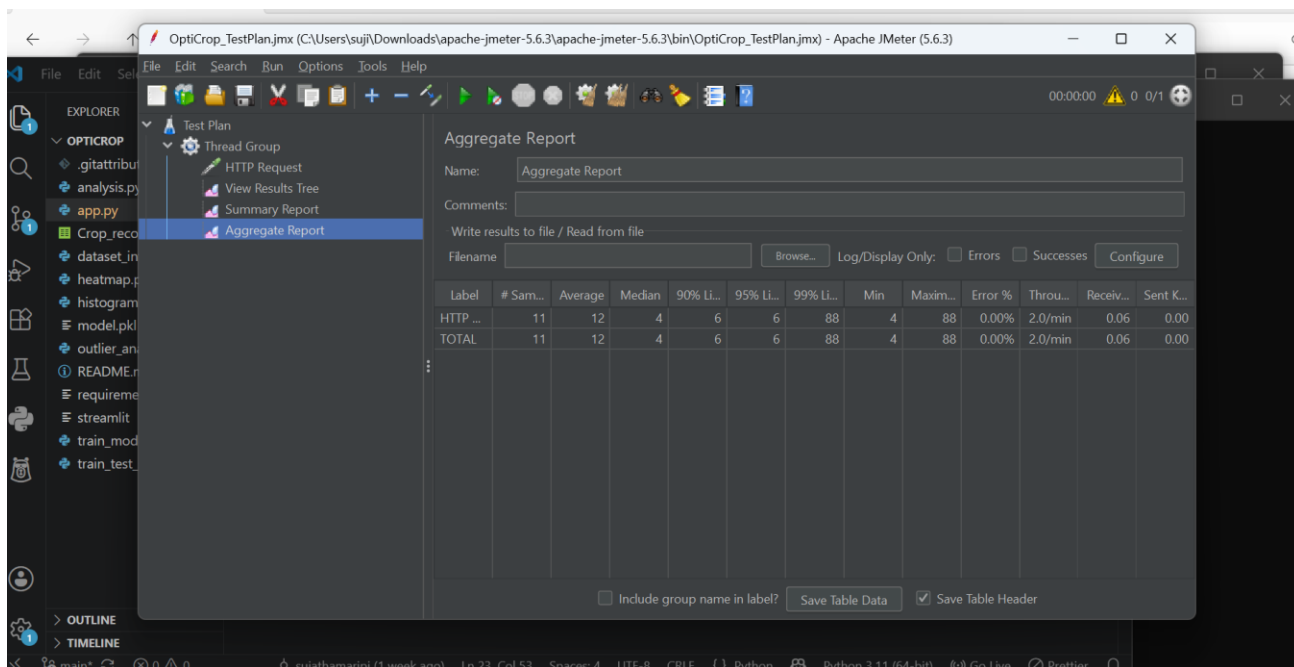


The screenshot shows the Apache JMeter 5.6.3 interface. The left sidebar displays the Explorer tree with the following structure:

- Test Plan
 - Thread Group
 - HTTP Request (selected)
 - View Results Tree
 - Summary Report
 - Aggregate Report

The main panel displays the 'View Results Tree' for the selected HTTP Request. The 'Text' tab is active, showing the following details:

- Name: View Results Tree
- Comments:
- Write results to file / Read from file:
 - Filename:
 - Browse...
 - Log/Display Only: ☐ Errors ☐ Successes
- Search: ☐ Case sensitive ☐ Regular exp.
- Sampler result:
 - Thread Name: Thread Group 1-1
 - Sample Start: 2026-07-10 15:09:19 IST
 - Load time: 4
 - Connect Time: 1
 - Latency: 4
 - Size in bytes: 1792
 - Sent bytes: 117
 - Headers size in bytes: 270
 - Body size in bytes: 1522
 - Sample Count: 1
 - Error Count: 0
 - Data type ("text"/"bin") : text
 - Response code: 200
 - Response message: OK



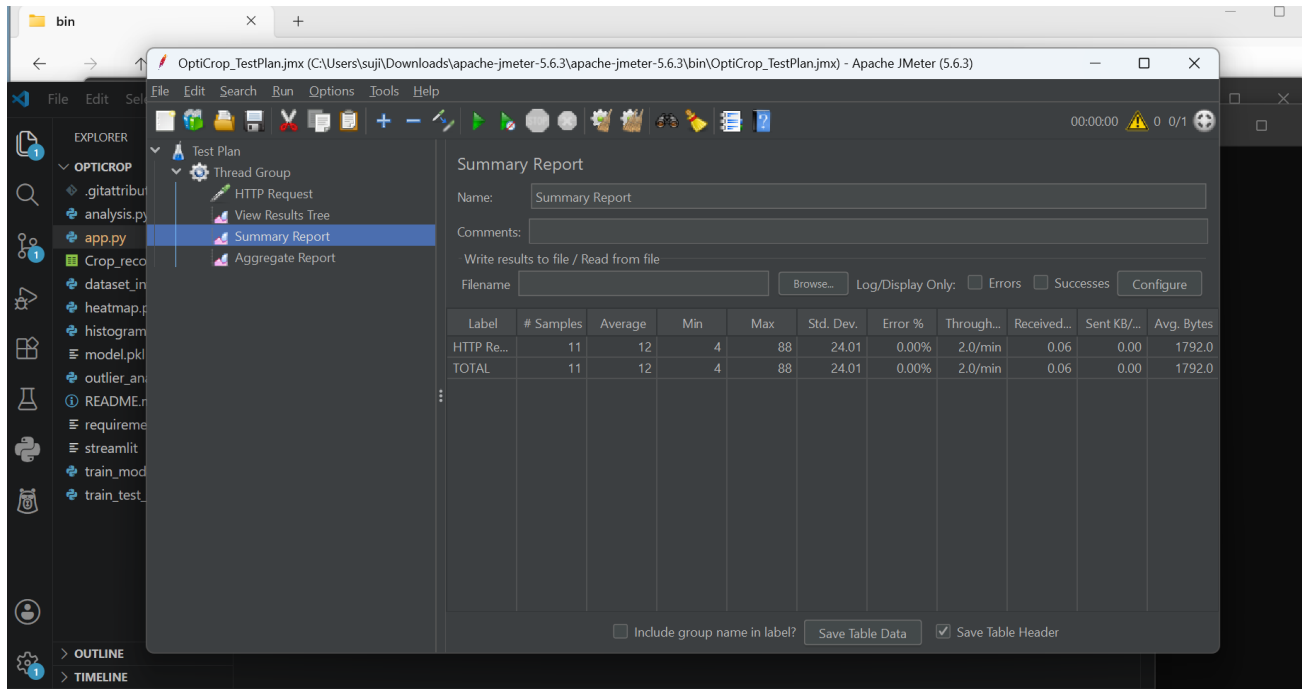
The screenshot shows the Apache JMeter 5.6.3 interface. The left sidebar displays the Explorer tree with the following structure:

- Test Plan
 - Thread Group
 - HTTP Request
 - View Results Tree
 - Summary Report
 - Aggregate Report (selected)

The main panel displays the 'Aggregate Report' for the selected HTTP Request. The 'Text' tab is active, showing the following details:

- Name: Aggregate Report
- Comments:
- Write results to file / Read from file:
 - Filename:
 - Browse...
 - Log/Display Only: ☐ Errors ☐ Successes
- Table:

Label	# Sam...	Average	Median	90% Li...	95% Li...	99% Li...	Min	Maxim...	Error %	Throu...	Receiv...	Sent K...
HTTP ...	11	12	4	6	6	88	4	88	0.00%	2.0/min	0.06	0.00
TOTAL	11	12	4	6	6	88	4	88	0.00%	2.0/min	0.06	0.00
- Include group name in label? ☐ Save Table Data ☒ Save Table Header



OptiCrop_TestPlan.jmx (C:\Users\sujit\Downloads\apache-jmeter-5.6.3\apache-jmeter-5.6.3\bin\OptiCrop_TestPlan.jmx) - Apache JMeter (5.6.3)

Summary Report

Name: Summary Report

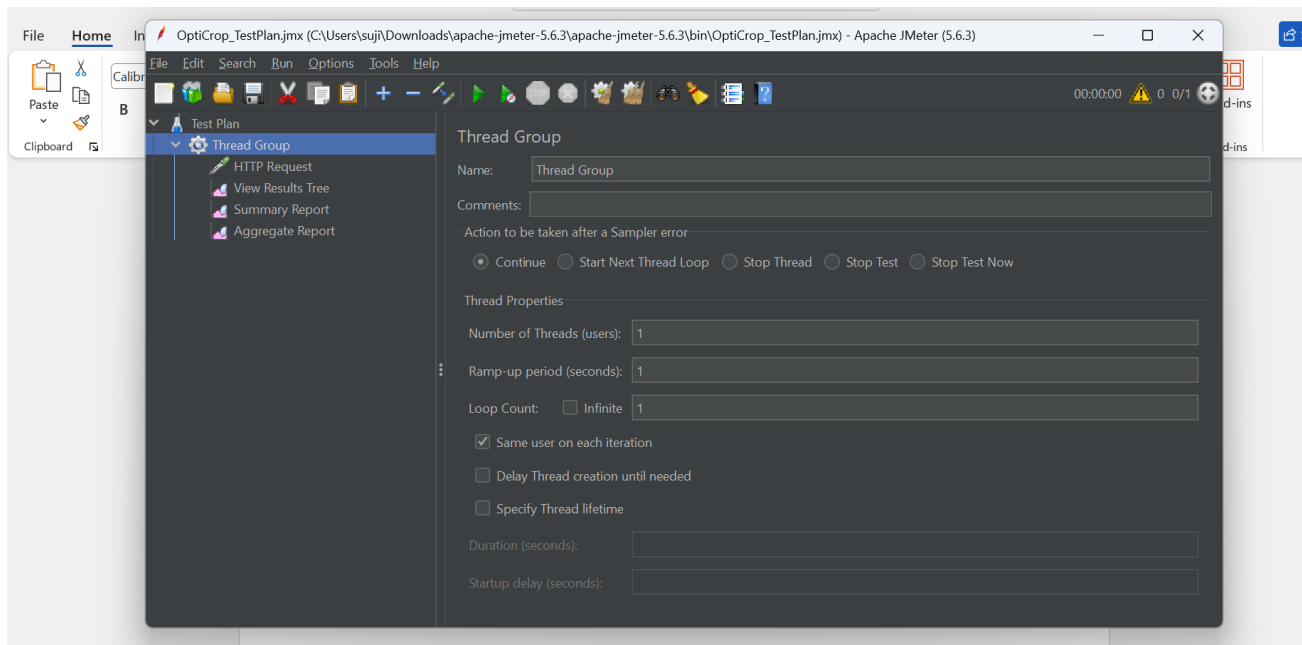
Comments:

Write results to file / Read from file

Filename: Browse... Log/Display Only: ☐ Errors ☐ Successes

Label	# Samples	Average	Min	Max	Std. Dev.	Error %	Through...	Received...	Sent KB/...	Avg. Bytes
HTTP Re...	11	12	4	88	24.01	0.00%	2.0/min	0.06	0.00	1792.0
TOTAL	11	12	4	88	24.01	0.00%	2.0/min	0.06	0.00	1792.0

☐ Include group name in label? ☒ Save Table Header



OptiCrop_TestPlan.jmx (C:\Users\sujit\Downloads\apache-jmeter-5.6.3\apache-jmeter-5.6.3\bin\OptiCrop_TestPlan.jmx) - Apache JMeter (5.6.3)

Thread Group

Name: Thread Group

Comments:

Action to be taken after a Sampler error

☒ Continue ☐ Start Next Thread Loop ☐ Stop Thread ☐ Stop Test ☐ Stop Test Now

Thread Properties

Number of Threads (users):

Ramp-up period (seconds):

Loop Count: ☐ Infinite

☒ Same user on each iteration

☐ Delay Thread creation until needed

☐ Specify Thread lifetime

Duration (seconds):

Startup delay (seconds):